

Math 2551 Worksheet: $\mathbb{R}^3$

1. Find the components of the vector in 3-space of length 3 lying in the yz -plane pointing upward at an angle of $\pi/6$ measured from the positive y -axis.
2. Which is traveling faster, a car whose velocity vector is $\langle 28, 33 \rangle$ km/h or a car whose velocity vector is $\langle 40, 0 \rangle$ km/h? At what speed is the faster car traveling?
3. Find the equation of a sphere of radius 4 in $\mathbb{R}^3$ centered at the point $(1, -2, 3)$.